Regularized Multivariate Two-way Functional Principal Component Analysis

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Outline

Introduction

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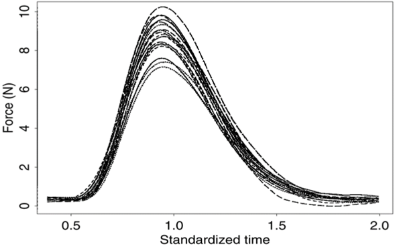

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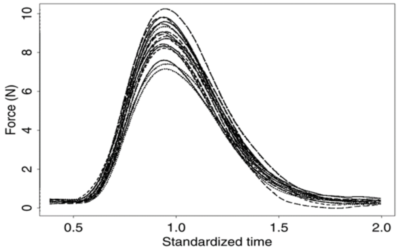

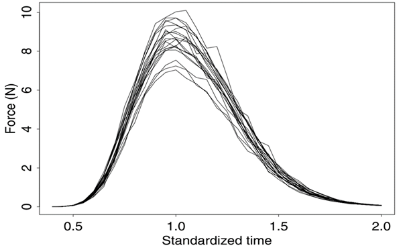

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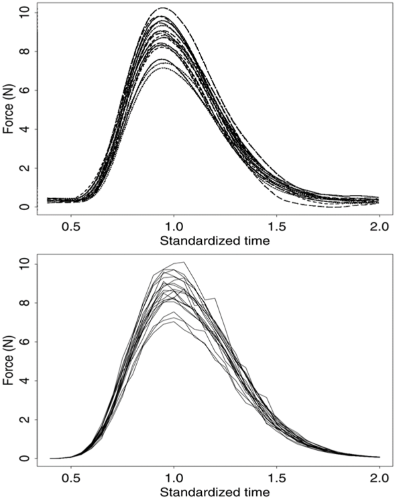
- These data are often recorded at discrete time points or grid locations, even though they originate from continuous processes in areas like engineering, finance, environmental science, and healthcare.
- Functional Data Analysis (FDA) is a statistical framework that treats these observations as realizations of smooth underlying functions, allowing for more accurate modeling and interpretation of continuous processes.



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- Impact: More adaptable, robust, and applicable across diverse scientific and business problems.

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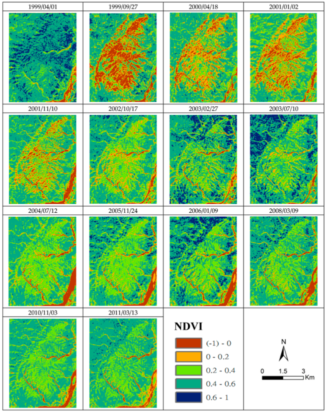

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  - Effective for dimension reduction in complex datasets.

Mathematical Foundation of FPCA

- **Goal:** Identify functional directions that maximize variance (low-rank approximation of functional data). For functional data $X \in \mathbb{R}^{n \times m}$ contains the discretized functional observations (rows correspond to subjects, columns to grid points), $v \in \mathbb{R}^m$ represents the estimated principal component (function), and $u \in \mathbb{R}^n$ denotes the associated principal component scores.

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- **Optimization steps:**

$$\text{Fix } v: \ u = \frac{Xv}{v^\top v} \qquad \text{and} \qquad \text{Fix } u: \ v = \frac{X^\top u}{u^\top u}$$

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- Algorithms: Based on iterative **power method** and **thresholding** updates.

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- Consider the SVD of X as $X = UDV^\top$, where U and V have orthonormal columns and D is diagonal with ordered singular values. In particular, for $X = udv^\top$, v is the first principal component and $u = ud$ gives the associated scores. With these representations, the power algorithm (described below) converges quickly, typically in only a few iterations.

Power Algorithm

Algorithm: Penalized Power Iteration

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- For notational convenience, we define $S(\alpha) = (I + \alpha\Omega)^{-1} \in \mathbb{R}^{m \times m}$, which simplifies expressions involving regularization. The penalty matrix Ω is set up so that larger values of the quadratic form $v^\top \Omega v$ mean rougher functions. This means that it penalizes functions that change quickly between time points.

Tunning Smoothness Parameters

- To select the optimal tuning parameters α efficiently, one can use a traditional Cross-Validation (CV) criterion and a computationally efficient closed-form Generalized Cross-Validation (GCV) criterion:

$$\text{CV}(\alpha) = \frac{1}{m}\sum_{j=1}^{m} \frac{\left[\{(I - S(\alpha))(X^T u)\}_{jj}\right]^2}{(1 - \{S(\alpha)\}_{jj})^2},$$

where $\{\cdot\}_{jj}$ denotes the j-th diagonal element.

$$\text{GCV}(\alpha) = \frac{1}{m} \frac{\|(I - S(\alpha))(X^T u)\|^2}{\left(1 - \frac{1}{m}\text{tr}\{S(\alpha)\}\right)^2}.$$

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- Sparse FPCA formulation [Shen and Huang, 2008]:

$$\min_{u,v} \|X - uv^\top\|_F^2 + p_\gamma(v)$$

where $p_\gamma(v)$ is a sparsity-inducing penalty.

Sparsity penalties

- **Lasso (Soft-thresholding):**

$$p_\gamma^{\text{soft}}(|\theta|) = 2\gamma|\theta|, \xrightarrow[\text{minimizer}]{} h_\gamma^{\text{soft}}(y) = \text{sign}(y)(|y|-\gamma)_+$$

- **Hard thresholding:**

$$p_\gamma^{\text{hard}}(|\theta|) = \gamma^2 I(|\theta| \neq 0), \xrightarrow[\text{minimizer}]{} h_\gamma^{\text{hard}}(y) = I(|y| > \gamma)\, y$$

- **SCAD penalty:**

$$p_\gamma^{\text{SCAD}}(|\theta|) = \begin{cases} 2\gamma|\theta|, & |\theta| \leq \gamma, \\ \dfrac{\theta^2 - 2a\gamma|\theta| + \gamma^2}{a-1}, & \gamma < |\theta| \leq a\gamma, \\ \dfrac{(a+1)\gamma^2}{2}, & |\theta| > a\gamma, \end{cases} \xrightarrow[\text{minimizer}]{} h_\gamma^{\text{SCAD}}(y) = \begin{cases} \text{sign}(y)(|y|-\gamma)_+, & |y| \leq 2\gamma, \\ \dfrac{(a-1)y - \text{sign}(y)a\gamma}{a-2}, & 2\gamma < |y| \leq a\gamma, \\ y, & |y| > a\gamma, \end{cases}$$

where $a = 3.7$ (Fan and Li [2001]).

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   3. **Normalize Right Singular Vector:** $v \leftarrow \dfrac{v}{\|v\|}$

Cross-Validation for Sparsity Selection

- **Sparsity parameter**: Tuning parameter controlling number of non-zero loadings in v ($0 =$ dense, $p =$ full sparsity).

Algorithm: K-fold CV Tuning Parameter Selection - Degree of sparsity

1. Randomly group the rows of side-by-side data matrix X into K roughly equal-sized groups, denoted as $X^1, \ldots, X^K$.

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   2. Calculate the K-fold CV scores defined as: (N is the number of grid points in X^k)

$$CV_j = \sum_{k=1}^{K} \frac{\|X^k - u^{-k}(j)v^k(j)\|^2}{N}$$

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3. Select the degree of sparsity as $j_0 = \arg\min\{CV(j)\}$.

Comparison & Tuning

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  - More stable & applicable to high-dimensional multivariate FDA

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- **Block structure example**:

$$X_i = \begin{bmatrix} x_{1i}(t_{i1}) & \cdots & x_{1i}(t_{i,m_i}) \\ \vdots & \ddots & \vdots \\ x_{ni}(t_{i1}) & \cdots & x_{ni}(t_{i,m_i}) \end{bmatrix}, \quad \mathbf{X} = [X_1 \; X_2 \; \ldots \; X_p]$$

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- The penalized reconstruction error is

$$\min_{u,v} \|\boldsymbol{X} - uv^\top\|_F^2 + \boldsymbol{\alpha}^\top (v^\top \boldsymbol{\Omega} v),$$

where $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_p)^\top$ controls smoothness.

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- The smoothing parameter $\boldsymbol{\alpha}$ is selected via generalized cross-validation (GCV), defined as

$$\mathrm{GCV}(\boldsymbol{\alpha}) = \frac{1}{M}\frac{\|(I - \boldsymbol{S}(\boldsymbol{\alpha}))(\mathbf{X}^T u)\|^2}{\left(1 - \frac{1}{M}\mathrm{tr}\{\mathbf{S}(\boldsymbol{\alpha})\}\right)^2}. \quad (1)$$

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- Algorithm CV Tuning for Sparsity and equation (1) are used to tune the sparsity level via K-fold CV and the smoothing parameter via GCV, respectively.

Simulation: Estimation Performance

- **Data-generating process:** Two functional variables:

$$X_{ij}^{(1)} = u_{i1} v_{11}(t_j) + u_{i2} v_{12}(t_j) + \epsilon_{ij}^{(1)}, \quad X_{ij}^{(2)} = u_{i1} v_{21}(t_j) + u_{i2} v_{22}(t_j) + \epsilon_{ij}^{(2)},$$

  - where $u_{i1} \sim N(0, \sigma_1^2)$, $u_{i2} \sim N(0, \sigma_2^2)$, $\epsilon_{ij}^{(k)} \sim N(0, \sigma^2)$, and $n = m = 101$, $t_j \in [-1, 1]$
- **True functional PCs:**
  - Variable 1: $v_{11}(t) = \frac{t + \sin(\pi t)}{s_1}, \qquad v_{12}(t) = \frac{\cos(3\pi t)}{s_2}$
  - Variable 2:

$$v_{21}(t) = \begin{cases} \dfrac{\sin(3\pi t)}{s_3}, & t \in (-\frac{1}{3}, \frac{1}{3}), \\ 0, & \text{otherwise}, \end{cases} \qquad v_{22}(t) = \begin{cases} \dfrac{\sin(2\pi t)}{s_4}, & t \leq -\frac{1}{3}, \\ \dfrac{\sin(\pi t)}{s_4}, & t \geq \frac{1}{3}, \\ 0, & \text{otherwise}. \end{cases}$$

Here, s_1, s_2, s_3, s_4 are normalizing constants ensuring unit L^2 norm.

Simulation: Estimation Performance

Scenarios tested:

1. Unpenalized Multivariate SVD (baseline)
2. Smoothed Multivariate SVD (smoothness penalty)
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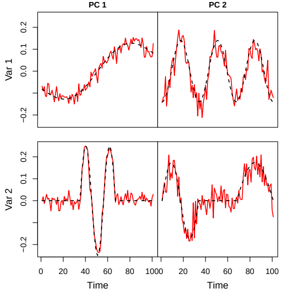

(a) MFPCA

Simulation: Estimation Performance

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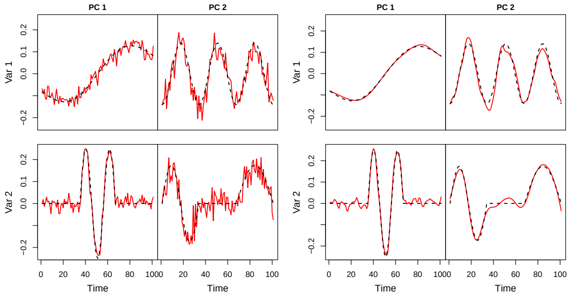

(a) MFPCA (b) Smoothed MFPCA

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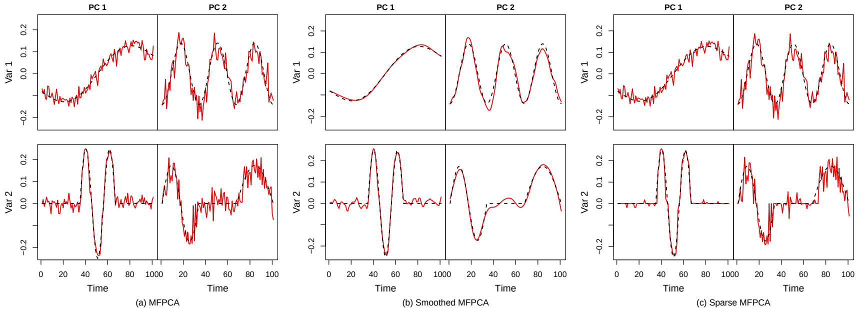

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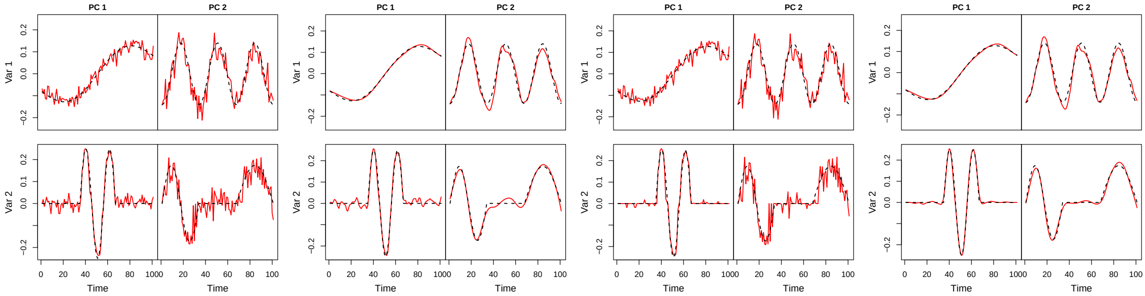


(a) MFPCA (b) Smoothed MFPCA (c) Sparse MFPCA (d) Smoothed and Sparse MFPCA

Simulation: Estimation Performance

Accuracy measures:

1. Variable-wise MSE:

$$\mathrm{MSE}_{k\ell} = \frac{1}{m}\sum_{j=1}^{m}(\hat{v}_{k\ell}(t_j) - v_{k\ell}(t_j))^2$$

2. Replication-averaged MSE:

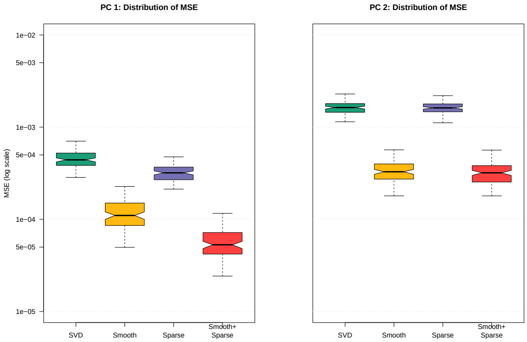
$$\overline{\mathrm{MSE}}_{k\ell} = \frac{1}{R}\sum_{r=1}^{R}\mathrm{MSE}_{k\ell}^{(r)}$$

3. Multivariate MSE:

$$\mathrm{MSE}_{\ell}^{(\mathrm{multi})} = \frac{1}{m}\sum_{j=1}^{m}\sum_{k=1}^{2}(\hat{v}_{k\ell}(t_j) - v_{k\ell}(t_j))^2$$

Simulation: Estimation Performance

- **Performance across four methods (SVD, Smooth, Sparse, Smooth+Sparse):**
  - Smoothness and/or sparsity **reduce MSE** compared to unregularized SVD.
  - **Smooth+Sparse** yields lowest error and most stable estimates.
  - Smooth estimator performs consistently well; sparsity alone less effective (esp. for PC2).
  - Joint regularization achieves best **bias–variance tradeoff**.



PC1: Quartiles and Mean log10(MSE)

Method	Q1	Median	Mean	Q3
SVD	-3.41	-3.35	-3.34	-3.28
Smooth	-4.07	-3.96	-3.92	-3.82
Sparse	-3.57	-3.50	-3.49	-3.43
Smooth+Sparse	**-4.38**	**-4.28**	**-4.22**	**-4.14**

PC2: Quartiles and Mean log10(MSE)

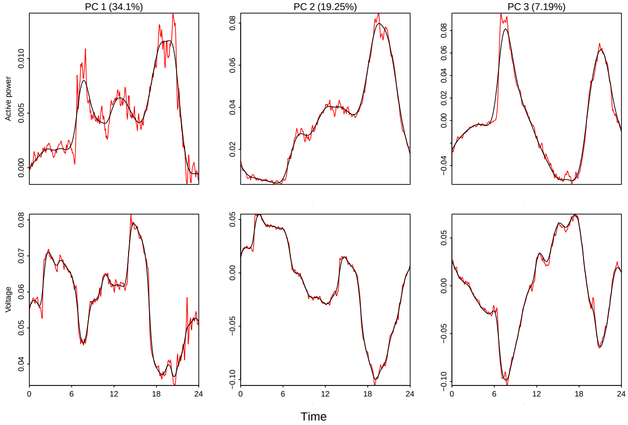
Method	Q1	Median	Mean	Q3
SVD	-2.84	-2.79	-2.79	-2.75
Smooth	-3.56	-3.48	-3.47	-3.40
Sparse	-2.83	-2.79	-2.79	-2.75
Smooth+Sparse	**-3.59**	**-3.50**	**-3.49**	**-3.42**

Application: Household Power Consumption

- **Dataset:** Bivariate functional data including active power and voltage consumption [Hebrail and Berard, 2012] for one household between December 2006 and November 2010.
- **Scaling:** To equalize the contribution of each variable in the multivariate analysis, we rescale them following [Happ and Greven, 2018].

$$\tilde{X}_j(t_i) = \hat{w}_j^{1/2} X_j(t_i),$$

$$\hat{w}_j = \left(\frac{1}{m}\sum_{i=1}^{m}\widehat{\mathrm{Var}}(X_j(t_i))\right)^{-1}.$$



First 3 PCs: MFPCA (red) vs ReMFPCA (black)

Regularization reduces noise while preserving the dominant daily consumption patterns, enhancing interpretability without losing key structure.

Two-way Regularized MFPCA

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  - Result: Low-rank, interpretable, noise-robust principal components for high-dimensional applications.

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- Multivariate v: $\boldsymbol{\Omega}_v = \text{blockdiag}(\Omega_1, \ldots, \Omega_p).$

Two-way Smoothed MFPCA: Conditional GCV

- Minimizers:

$$u = \frac{\boldsymbol{S}_u(\alpha_u)\boldsymbol{X}v}{v^\top(\boldsymbol{I}+\boldsymbol{\alpha}_v\boldsymbol{\Omega}_v)\,v} = \frac{\boldsymbol{S}_u(\alpha_u)}{1+\boldsymbol{\alpha}_v\boldsymbol{R}_v(v)}\,\frac{\boldsymbol{X}v}{\|v\|^2}, \qquad v = \frac{\boldsymbol{S}_v(\boldsymbol{\alpha}_v)\boldsymbol{X}^\top u}{u^\top(\boldsymbol{I}+\alpha_u\boldsymbol{\Omega}_u)\,u} = \frac{\boldsymbol{S}_v(\boldsymbol{\alpha}_v)}{1+\alpha_u\boldsymbol{R}_u(u)}\,\frac{\boldsymbol{X}^\top u}{\|u\|^2}.$$

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- **Optimization:** Alternate updates of u and v using GCV until convergence $\to$ two-way regularized components.

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- **Result:** Robust, interpretable, noise-resistant components for high-dimensional data.

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- Data matrix $\boldsymbol{X}$: seek u, v solving:

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- Penalties: smoothness, sparsity (soft/hard thresholding, SCAD, group lasso, etc.).

Algorithm: Two-way Smooth + Sparse MFPCA (Sequential Power)

1. **Initialization:** Rank-one SVD of $\boldsymbol{X}$: $\boldsymbol{X} \approx s\, u^{(0)} v^{(0)\top}$; set $u \leftarrow s\, u^{(0)}$, $v \leftarrow v^{(0)}$.
2. **Repeat until convergence:**
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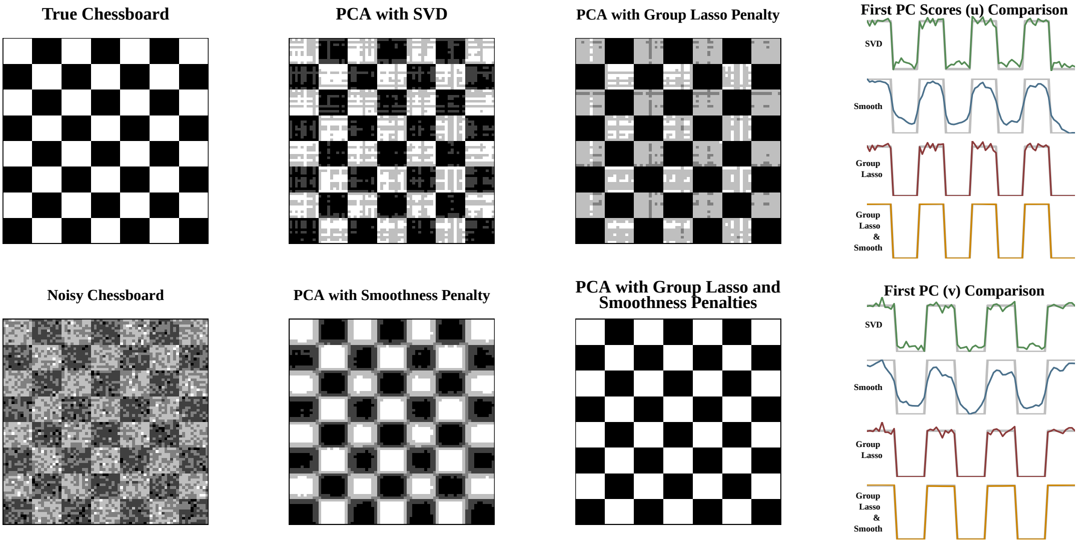
   3. $v \leftarrow v/\|v\|$
3. $\boldsymbol{X} \leftarrow \boldsymbol{X} - \sigma\, u\, v^\top$ to extract the next component.

Chess Board


True Chessboard
PCA with SVD
PCA with Group Lasso Penalty
First PC Scores (u) Comparison
SVD
Smooth
Group Lasso
Group Lasso & Smooth
Noisy Chessboard
PCA with Smoothness Penalty
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First PC (v) Comparison
SVD
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Selection of Regularization Parameters

- Four sets of tuning parameters:
  - α_u: smoothness of u, γ_u: sparsity of u
  - $\boldsymbol{\alpha}_v$: smoothness of v, $\boldsymbol{\gamma}_v$: sparsity of v
- **Challenge:** Ordering of tuning (smoothness vs. sparsity) affects convergence and solutions.
- **Strategy: Conditional tuning**
  1. Initialize all penalties at 0.
  2. Tune γ_u via K-fold CV.
  3. Sequentially tune $\gamma_{v,i}$ using Algorithm~**??**.
  4. With sparsity fixed, tune α_u by GCV.
  5. Tune $\alpha_{v,i}$ using two-way GCV.
  6. Iterate steps 2–5 until stable.
- This alternating scheme **isolates sparsity vs. smoothness** while ensuring accuracy + interpretability.

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1. Split $\boldsymbol{X} \in \mathbb{R}^{n\times M}$ into K column groups $\{\boldsymbol{X}^{(1)}, \ldots, \boldsymbol{X}^{(K)}\}$.

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Variance Explained: Classical vs Regularized FPCA

- **Classical FPCA:** Loadings v_j orthonormal; scores $u_j = Xv_j$ uncorrelated. Variance explained by first J PCs:

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- **Issue under regularization:** smoothness/sparsity break orthogonality $\rightarrow$ scores become correlated $\rightarrow$ naive sum $\sum \|u_j\|^2$ **double-counts** variance (cf. Huang et al., 2008).

Subspace-Projection Definition of Explained Variance

- Normalize loadings and stack:

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- Projected data and explained variance:

$$X_J = XH_J, \qquad \mathrm{V}_{\mathrm{tot}} = \mathrm{tr}(X^\top X), \qquad \mathcal{V}_J = \|X_J\|_F^2 = \mathrm{tr}\big(H_J X^\top X H_J\big).$$

PVE, Incremental PVE, and Properties

- Incremental variance:

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- **Key properties:**
  - No double-counting (works with correlated scores).
  - Reduces to classical PCA when $V_J^\top V_J = I_J$.
  - Monotone in J ($\mathcal{V}_J$ increases).
  - $\Delta\mathcal{V}_j =$ unique variance added by component j.

Simulation: Two-way Functional Data

- **Data-generating process:**
  $X_{ij}^{(1)} = u_{i1}v_{11}(t_j) + u_{i2}v_{12}(t_j) + \epsilon_{ij}^{(1)}, \quad X_{ij}^{(2)} = u_{i1}v_{21}(t_j) + u_{i2}v_{22}(t_j) + \epsilon_{ij}^{(2)},$
- **Latent scores:** generated as smooth functions

$$u_1(s) = \begin{cases} \sin(\pi s), & s > 0, \\ 0, & \text{otherwise}, \end{cases} \qquad u_2(s) = \sin(2\pi s), \quad s \in [-1, 1].$$

- **Functional PCs:**
  - Variable 1: $v_{11}(t) = \frac{t+\sin(\pi t)}{s_1}, \qquad v_{12}(t) = \frac{\cos(3\pi t)}{s_2}$
  - Variable 2:

$$v_{21}(t) = \begin{cases} \dfrac{\sin(3\pi t)}{s_3}, & t \in (-\frac{1}{3}, \frac{1}{3}), \\ 0, & \text{otherwise}, \end{cases} \qquad v_{22}(t) = \begin{cases} \dfrac{\sin(2\pi t)}{s_4}, & t \leq -\frac{1}{3}, \\ \dfrac{\sin(\pi t)}{s_4}, & t \geq \frac{1}{3}, \\ 0, & \text{otherwise}. \end{cases}$$

Evaluation Metrics

- **Integrated Squared Error (ISE):**
  For replicate r and component u_1:

$$\mathrm{ISE}_r^{(u_1,\text{method})} = \frac{1}{m}\sum_{j=1}^{m}\Big(u_1(t_j) - \widehat{u}_1^{(\text{method})}(t_j)\Big)^2.$$

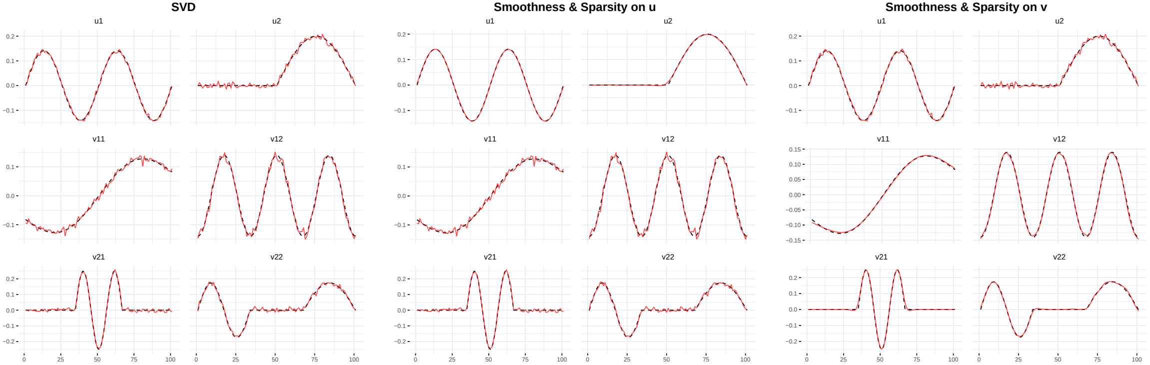
- **Relative ISE (R_ISE):** ratio vs best method

$$R_r^{(u_1,\text{method})} = \frac{\mathrm{ISE}_r^{(u_1,\text{method})}}{\mathrm{ISE}_r^{(u_1,\text{best})}}.$$

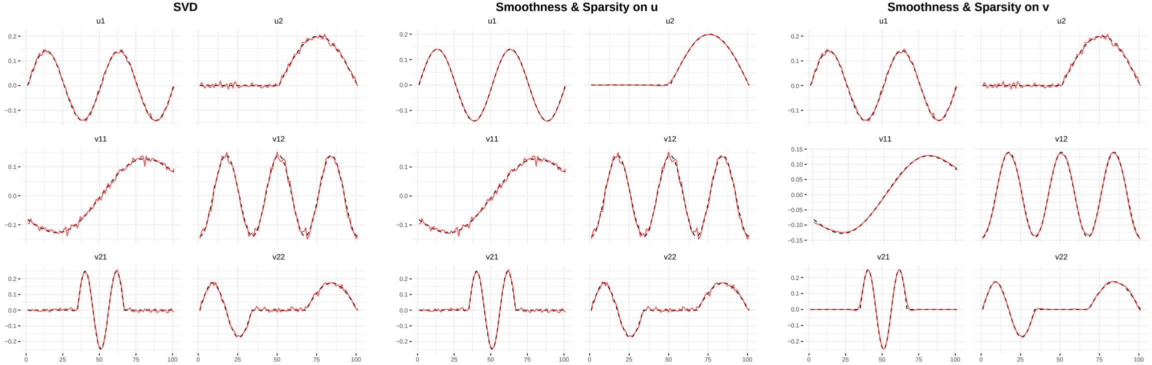
- **Monte Carlo averages:**

$$\overline{R}^{(u_1,\text{method})} = \frac{1}{N}\sum_{r=1}^{N} R_r^{(u_1,\text{method})}, \qquad \mathrm{SE}(\overline{R}) = \sqrt{\frac{1}{N(N-1)}\sum_{r=1}^{N}\Big(R_r - \overline{R}\Big)^2}.$$

Simulation Results



Simulation Results



Simulation Results

Table 3: Mean ISE for each method and parameter

Method	u1	u2	v1	v2
SVD	0.3651	0.1587	0.00005	0.00010
Smooth+Sparse v	0.3651	0.1587	**0.00002**	**0.00002**
Smooth+Sparse u	**0.3650**	**0.1584**	0.00005	0.00010
Two-way Smoothness	**0.3650**	0.1585	**0.00002**	**0.00002**
Two-way Sparsity	0.3651	0.1586	0.00004	0.00009
Two-way Sm+Sp	**0.3650**	**0.1584**	**0.00002**	**0.00002**

Table 4: Mean Relative ISE for each method and parameter

Method	u1	u2	v1	v2
SVD	1.000	1.001	8.21	5.23
Two-way Sparsity	1.000	1.001	7.71	4.61
Smooth+Sparse v	1.000	1.001	1.05	**1.01**
Smooth+Sparse u	**1.000**	**1.000**	8.19	5.21
Two-way Smoothness	**1.000**	**1.000**	**1.00**	1.21

- Two-way Smooth+Sparse consistently yields the lowest errors across u and v.

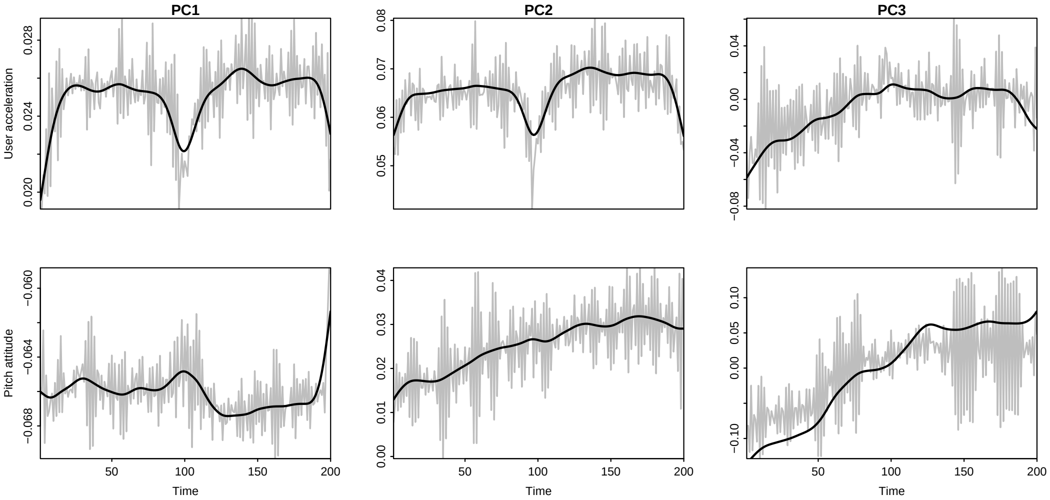
Application: Motion Sense Data

- **Dataset:** Acceleration and pitch from 24 people, 4 activities (jogging, walking, sitting, standing), about 2–3 min each.
- **Goal:** Compare SVD vs **two-way sparse + smooth ReMFPCA** on these multivariate functional signals.
- **Rescaling (Happ & Greven, 2018):** balance variables so each contributes equally.

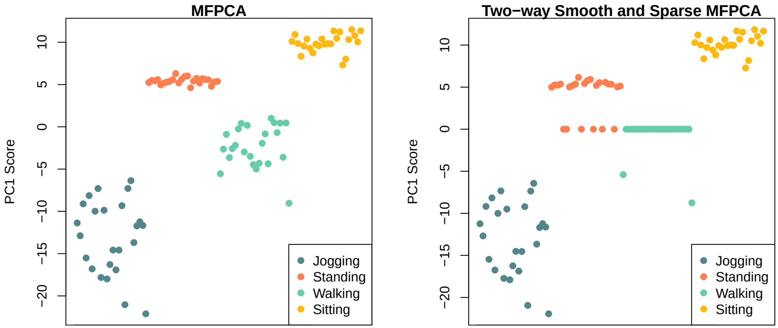
$$\hat{w}_j = \left(\frac{1}{m} \sum_{i=1}^{m} \widehat{\mathrm{Var}}(X_j(t_i)) \right)^{-1}, \qquad \tilde{X}_j(t_i) = \hat{w}_j^{1/2} X_j(t_i).$$

- **Penalties used:** smoothness + sparsity on loadings v; sparsity on scores u (treated as random effects).

Results: Functional PCs (SVD vs ReMFPCA)



Results: PC Scores and Interpretation



- Sparsity on scores: **PC1 scores for walking about 0** (partially standing too).
- Interpretation: walking contributes little to PC1; removing it **improves interpretability** without hurting fit.
- Takeaway: **Two-way smooth + sparse ReMFPCA** yields cleaner PCs and activity-informative scores.

Conclusion & Future Work

- **Unified FPCA Framework**
  - Combines **smoothness** (denoise, interpretability) + **sparsity** (variable selection).
  - Extends from **univariate** $\rightarrow$ **multivariate** $\rightarrow$ **two-way functional data**.
- **Methodology**
  - Penalized SVD with roughness + ℓ_1 penalties.
  - Two-way regularization: smoothness & sparsity on both **scores (*u*)** and **loadings (*v*)**.
  - Efficient parameter tuning: **conditional GCV** & **K-fold CV** (with 1-SE rule).
- **Results**
  - Simulations & applications (mortality, call-center, image data).
  - Outperforms one-way or single-penalty methods.
  - Produces **low-rank, denoised, interpretable components**.

Accessible Implementation: R Package & Future Work

- Implemented in **R package `ReMPCA` (GitHub)**
  - Univariate & multivariate FPCA with penalties.
  - Two-way MFPCA for matrix-valued functions.
  - Automated tuning (CV, GCV, 1-SE rule).
  - Diagnostic tools: variance explained, visualization, heatmaps.
  - Early support for **hybrid data (scalar + functional + image)**.
- **Hybrid Data Extensions**
  - Image–Functional Hybrid PCA $\rightarrow$ simultaneous dimension reduction.
  - Scalar–Functional Integration $\rightarrow$ joint low-dim space.
  - Nonlinear Extensions $\rightarrow$ kernel FPCA, neural nets.
- **Applications:**
  - Neuroimaging
  - Personalized medicine
  - Environmental monitoring

  Takeaway: Smooth + sparse + two-way FPCA offers a **theoretical foundation, practical algorithms, and open software** to enable next-generation functional data analysis.

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